

PKCERT AI & Software Development Internship

Task 21 – Regularization Techniques in Deep Learning

Objective:

Develop practical skills in improving deep learning model performance using regularization techniques including Dropout, Batch Normalization, and Early Stopping.

Total Marks: 100

Part A – Dataset Selection & Preparation (20 Marks)

1. Select a public dataset for a deep learning classification problem.
2. Describe the dataset, features, and target variable.
3. Preprocess the data and split it into training, validation, and testing sets.

Part B – Model Development (30 Marks)

1. Develop a baseline deep learning model.
2. Train and evaluate the model.
3. Document the model architecture and training process.

Part C – Regularization Techniques (35 Marks)

1. Apply Dropout and evaluate its impact.
2. Apply Batch Normalization and analyze performance improvements.
3. Implement Early Stopping and compare results with the baseline model.
4. Compare models using Accuracy, Precision, Recall, F1-Score, Loss Curves, and Confusion Matrix.

Part D – Comparative Analysis & Documentation (15 Marks)

1. Compare all approaches.
2. Discuss advantages, limitations, and recommend the most suitable regularization technique based on results.

Assessment Area	Marks
Dataset Selection & Preparation	20
Model Development	30
Regularization Techniques	35
Comparative Analysis & Documentation	15

Total	100
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